

Lesson 1: Introduction to Probability

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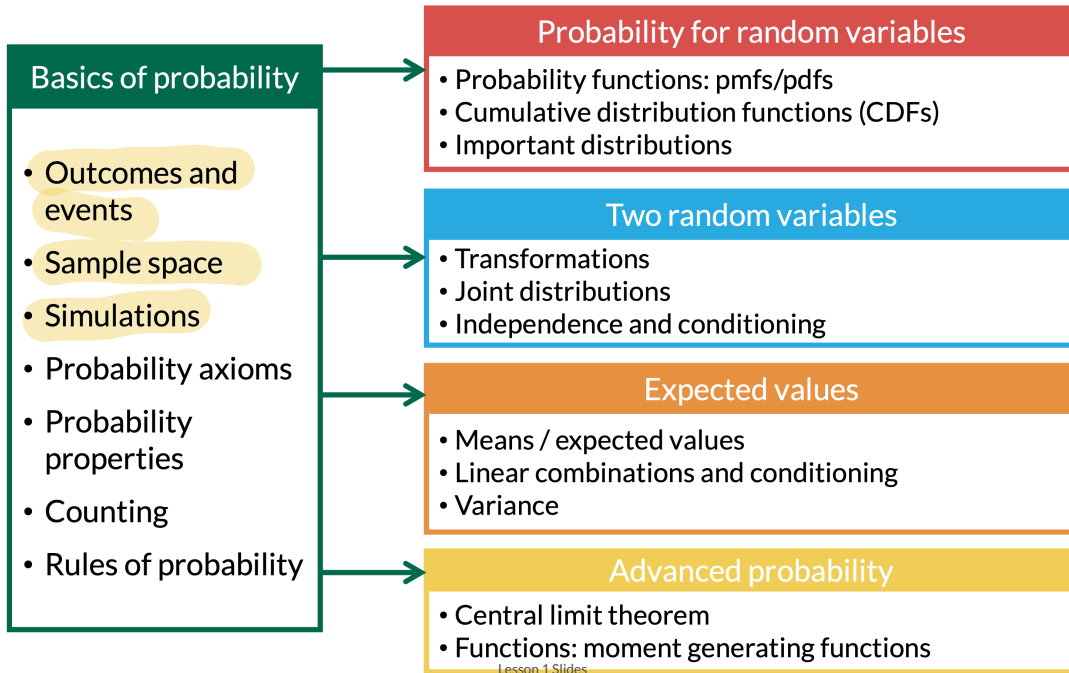
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Learning Objectives

1. Use simulations (e.g., coin flips in R) to illustrate randomness in equally likely outcomes.
2. Identify sample spaces and events for basic experiments, and define probability calculations.

Where are we?

- Big part of this lesson is **motivating** why we care about probability and simulations, then we'll dive into **some basics**



~~What is probability? (1/2)~~ Probability IRL

- We hear the word “probability” pretty often
 - Common in news reports, advertisements, sports, medicine, etc.

Researchers say the *probability* of living past 110 is on the rise

CNBC, 2021

- We may hear “probability” or similar words like “chance,” “likelihood,” or “odds”

Scientists fine-tune *odds* of asteroid Bennu hitting Earth through 2300 with NASA probe’s help

Space.com, 2021

Before we take one more step

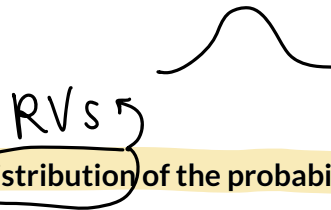
- The following few slides *use some undefined words* to **define new words** that in turn *define the previously undefined words*
- It's confusing!
- We're going off assumption that we all have some daily understanding of probability, so stop me if you are confused!

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Probability and randomness

- Probability requires **randomness**
- Something is **random** if there are many potential **outcomes**, but there is **uncertainty** which **outcome** will occur
- **Outcomes** can be “**equally likely**,” meaning each **outcome** has the *same* probability of happening
 - But **random** does **NOT** necessarily mean **equally likely**
- We often use **physical randomness** to demonstrate **equally likely** outcomes
 - Think: **flipping coins, rolling a dice, drawing cards**
- The occurrence of outcomes can be **uncertain**, but there is an **underlying distribution of the probability** of outcomes
 - There is a **distribution of outcomes over large number of (hypothetical repetitions)**



A single coin flip then 100 coin flips

Seeing Theory, Chapter 1: Basic Probability, Chance Events

This site is archived for reference

English

Seeing Theory

Chapter 1

Basic Probability

This chapter is an introduction to the basic concepts of probability theory.



Chance Events

Expectation

Variance

How do we simulate this in R?

- We know that heads and tails are **equally likely** for a single flip

```
1 set.seed(13) hold our space as we jump into R's random pool
2 coin <- c("heads", "tails")
3 sample(coin, 1)
[1] "tails"
```

- When we only flip the coin once, we only get one outcome (heads or tails)
 - We cannot see any distribution of the probability of outcomes



What if I sample 10 coin flips?

- How many heads do we get? Are we getting closer to a distribution of heads and tails that we expect?

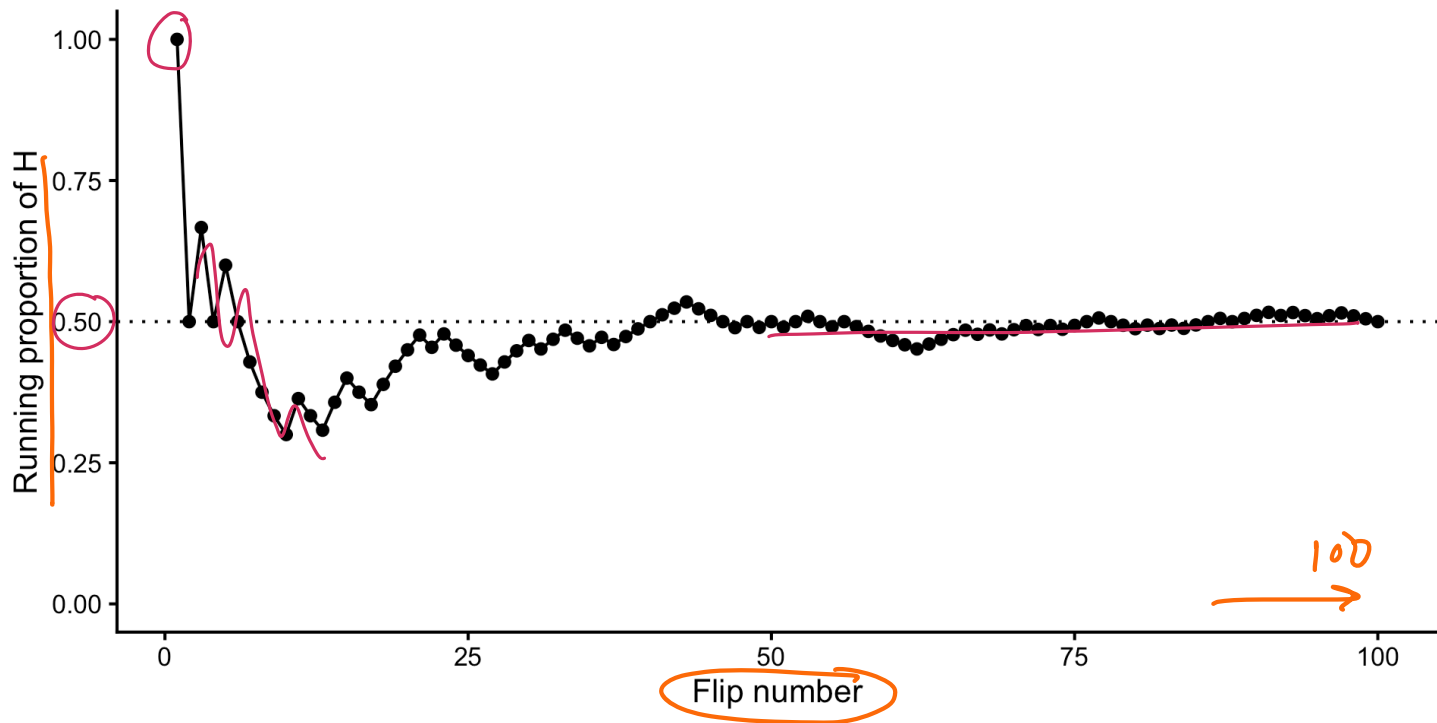
Results and running proportion of H for 10 flips of a fair coin.

Flip	Result	Running count of H	Running proportion of H
1	H	1	1.000
2	T	1	0.500
3	H	2	0.667
4	T	2	0.500
5	H	3	0.600
6	T	3	0.500
7	T	3	0.429
8	T	3	0.375
9	T	3	0.333
10	T	3	0.300

∞ ↓

↓ 0.5

What if I sample 100 coin flips?



What's the point?

- We know the probability of a heads is 0.5!
- Why do we need to simulate 100s or 1000s of coin flips?
 - With enough repetitions, we can ^{true} use simulations to approximate the probability of an event
- Okay, but why is that helpful? (next slide!)

Why are simulations important? (bigger picture)

In the previous example, coding a simulation seemed more educational than necessary

1. Simulations can help us (or be necessary) to **solve a problem when calculations are complex**
 - We can often calculate probabilities mathematically, but we will eventually get to complex calculations
2. Simulations are a great way to **check your work!**
3. Simulation based **reasoning** is helpful in statistics
 - You'll see this in confidence intervals in Biostatistics courses
4. Simulations allow you to **change assumptions easily** and see how they affect your results
5. It is often **how statisticians "run" experiments** on their methods of hypotheses

BSTA 511

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Outcomes, events, sample spaces

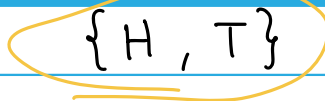
Definition: Outcome

The possible results in a random phenomenon.



Definition: Sample Space

The **sample space** S is the set of *all* outcomes



Definition: Event

An **event** is a *collection of some* outcomes. An event can include multiple outcomes or no outcomes (a subset of the sample space).

$$\begin{array}{ccccccc} H & T & \emptyset & \{H, T\} & P(S) = \\ P(H) = 0.5 & P(T) = 0.5 & P(\emptyset) = 0 & \underline{1} & \end{array}$$

When thinking about events, think about outcomes that you might be asking the probability of. For example, what is the probability that you get a heads or a tails in one flip? (Answer: 1)

Coin Toss Example: 1 coin (1/3)

Single coin toss

Suppose you toss one coin.

- What are the possible outcomes?
- What is the sample space?
- What are the possible events?

Coin Toss Example: 1 coin (2/3)

Suppose you toss one coin.

- What are the possible outcomes?

- Heads (H)

- Tails (T)

Note

When something happens at random, such as a coin toss, there are several possible outcomes, and exactly one of the outcomes will occur.

Coin Toss Example: 1 coin (3/3)

- What is the sample space?

$$S = \{ \underline{H}, \underline{T} \}$$

- What are the possible events?

- H

- T

- $\{H, T\}$

- $\emptyset$ (empty set)

if H in flip
or T in flip,
meet requirement of
this event

$$|S| = 2$$

$$2^{|S|} = 2^2$$

$$= 4$$

Note #1

We use curly brackets ($\{\}$) to denote a set (collecting a list of outcomes or values)

Note #2

The total number of possible events is

$$2^{|S|}$$

where $|S|$ is the total number of outcomes in the sample space. Also, possible events are not necessarily something that can actually occur (i.e. getting a heads and a tails on a single coin flip)

Coin Toss Example: 2 coins

Suppose you toss two coins.

- What is the sample space? Assume the coins are distinguishable

$$S = \{ HH, HT, TH, TT \}$$

4 outcomes in sample space

- What are some possible events?

$$A = \text{exactly one } H = \{ HT, TH \}$$

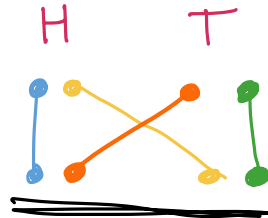
$$\cancel{B} = \text{at least one } H = \{ HH, HT, TH \}$$

■

■

→ coin 1

→ coin 2



poss events:

$$2^{151}$$

$$2^4 = 16$$

More info on events and sample spaces

- We usually use capital letters from the beginning of the alphabet to denote events. However, other letters might be chosen to be more descriptive.

▪ Examples: A, B, C, A_1, A_2

- We can also define a new event as a combination of other events (next lesson)

▪ Examples: $A \cup B$ (union), $A \cap B$ (intersection), A^C (complement)

$$\hookrightarrow \underline{C} = \underline{A \cup B}$$

- We use the notation $|S|$ to denote the size of the sample space.

$$\underline{|A|} = \# \text{ of outcomes in event } A$$

- The total number of possible events is $2^{|S|}$, which is the total number of possible subsets of S .

- The **empty set**, denoted by $\emptyset$, is the set containing no outcomes.

$$\emptyset$$

Let's define probability with events and spaces

If S is a finite sample space, with **equally likely** outcomes, then

$$\mathbb{P}(A) = \frac{|A|}{|S|} \rightarrow \begin{array}{l} \text{size of } A \\ \text{size of sample space} \end{array}$$

In human speak:

- For **equally likely** outcomes, the probability that a certain event occurs is: the number of outcomes within the event of interest ($|A|$) **divided by** the total number of possible outcomes ($|S|$)

$$\mathbb{P}(A) = \frac{\text{total number of outcomes in event } A}{\text{total number of outcomes in sample space}}$$

- Thus, it is important to be able to count the outcomes within an event

A probability is a function...

- $\mathbb{P}(A)$ is a function with
 - **Input:** event A from the sample space S , ($A \subseteq S$)
 - $A \subseteq S$ means “A contained within S ” or “A is a subset of S ”
 - **Output:** a number between 0 and 1 (inclusive)

↳ probability

- The **probability function** maps an event (input) to value between 0 and 1 (output)
 - When we speak of the probability function, we often call the values between 0 and 1 “probabilities”
 - Example: “The probability of drawing a heart is 0.25” for $P(\text{heart}) = 0.25$

- The probability function needs to follow some specific rules (called axioms)!

$$P(A) = \frac{|A|}{|S|}$$

$$\% = 100\% \times P(A)$$

Coin Toss Example Revisited: 2 coins

- From our sample space and events:

- Sample space: $S = \{HH, HT, TH, TT\}$ ← 4

→ Event A: $A = \underline{\text{exactly one H}} = \{HT, TH\}$

- Event B: $B = \underline{\text{at least one H}} = \{HT, TH, HH\}$

- Calculate $P(A)$ and $P(B)$

$$P(A) = \frac{|A|}{|S|} = \frac{2}{4} = \frac{1}{2}$$

$$P(B) = \frac{|B|}{|S|} = \frac{3}{4}$$

$$P(A) + P(B) \neq 1$$

sum of prob of events
do NOT need to be
1 (or less than 1)

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